

# resolve\_cross\_validation\_strategy.R

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| resolve_cross_validation_strategy   |  |
| Assess Cross-Validation Feasibility |  |

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## Description

Evaluates full-model and candidate training-partition diagnostics and selects the first viable cross-validation strategy.

## Usage

```
resolve_cross_validation_strategy(  
  data_partition_diagnostics = NULL,  
  min_train_locations = NULL,  
  min_train_samples = NULL,  
  min_train_taxa = NULL,  
  min_mem_locations = NULL  
)
```

## Arguments

`data_partition_diagnostics`  
Data frame with one row for the full model and one row per candidate training partition. Required columns are 'cv\_strategy', 'repeat\_id', 'effective\_folds', 'fold\_id', 'n\_train\_locations', 'n\_train\_samples', 'n\_train\_taxa', and 'n\_train\_mem\_locations'. Supported strategies are "full\_model", "spatially\_stratified\_group\_kfold", and "leave\_one\_location\_out".

`min_train_locations`, `min_train_samples`, `min_train_taxa`  
Positive integer thresholds required in the full model and every selected training partition.

`min_mem_locations`  
Positive integer minimum number of locations required for MEM construction.

**Details**

The full model is checked first. Viable grouped candidates are preferred in ascending fold-count order, followed by leave-one-location-out. A viable full model without any viable holdout requires tier-pooled regularization.

**Value**

One-row tibble containing full-data counts, feasibility flags, selected 'cv\_strategy', 'effective\_folds', and 'cv\_feasibility\_status'.

**Examples**

```
data_diagnostics <-  
  tibble::tibble(  
    cv_strategy = "full_model",  
    repeat_id = 0L,  
    effective_folds = NA_integer_,  
    fold_id = 0L,  
    n_train_locations = 7L,  
    n_train_samples = 70L,  
    n_train_taxa = 10L,  
    n_train_mem_locations = 7L  
  )  
resolve_cross_validation_strategy(  
  data_partition_diagnostics = data_diagnostics,  
  min_train_locations = 5L,  
  min_train_samples = 5L,  
  min_train_taxa = 3L,  
  min_mem_locations = 4L  
)
```

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